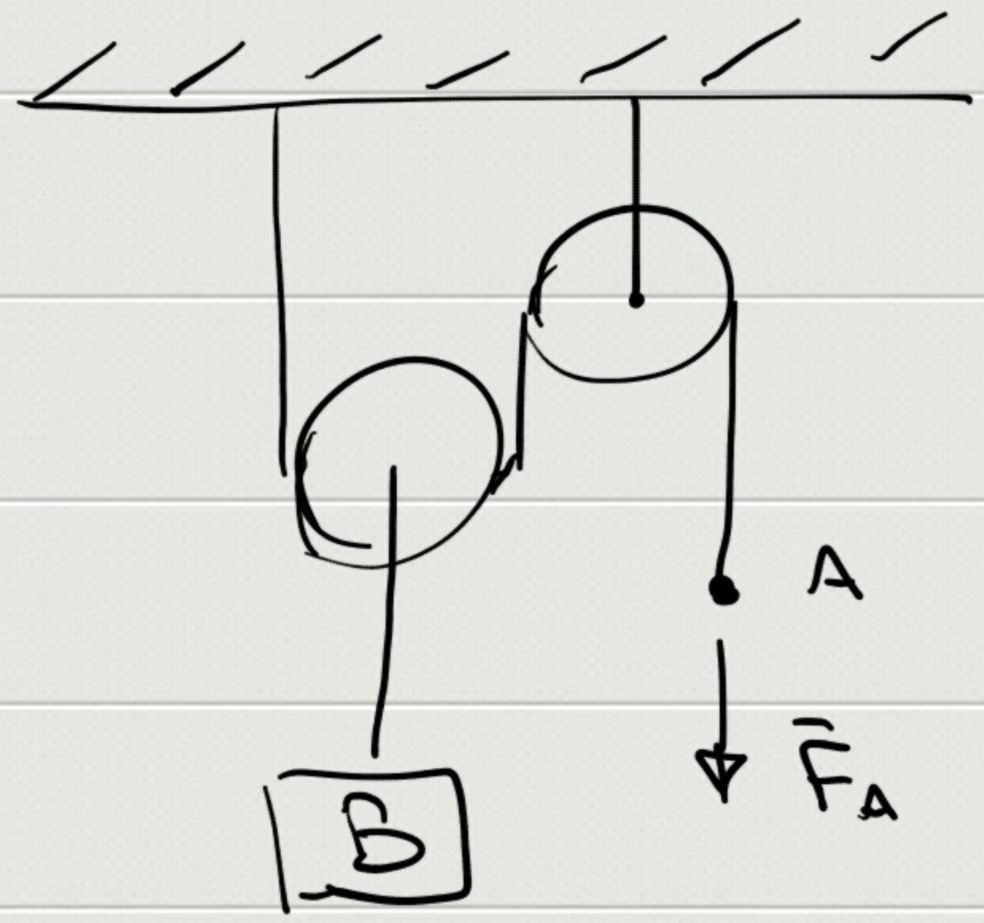
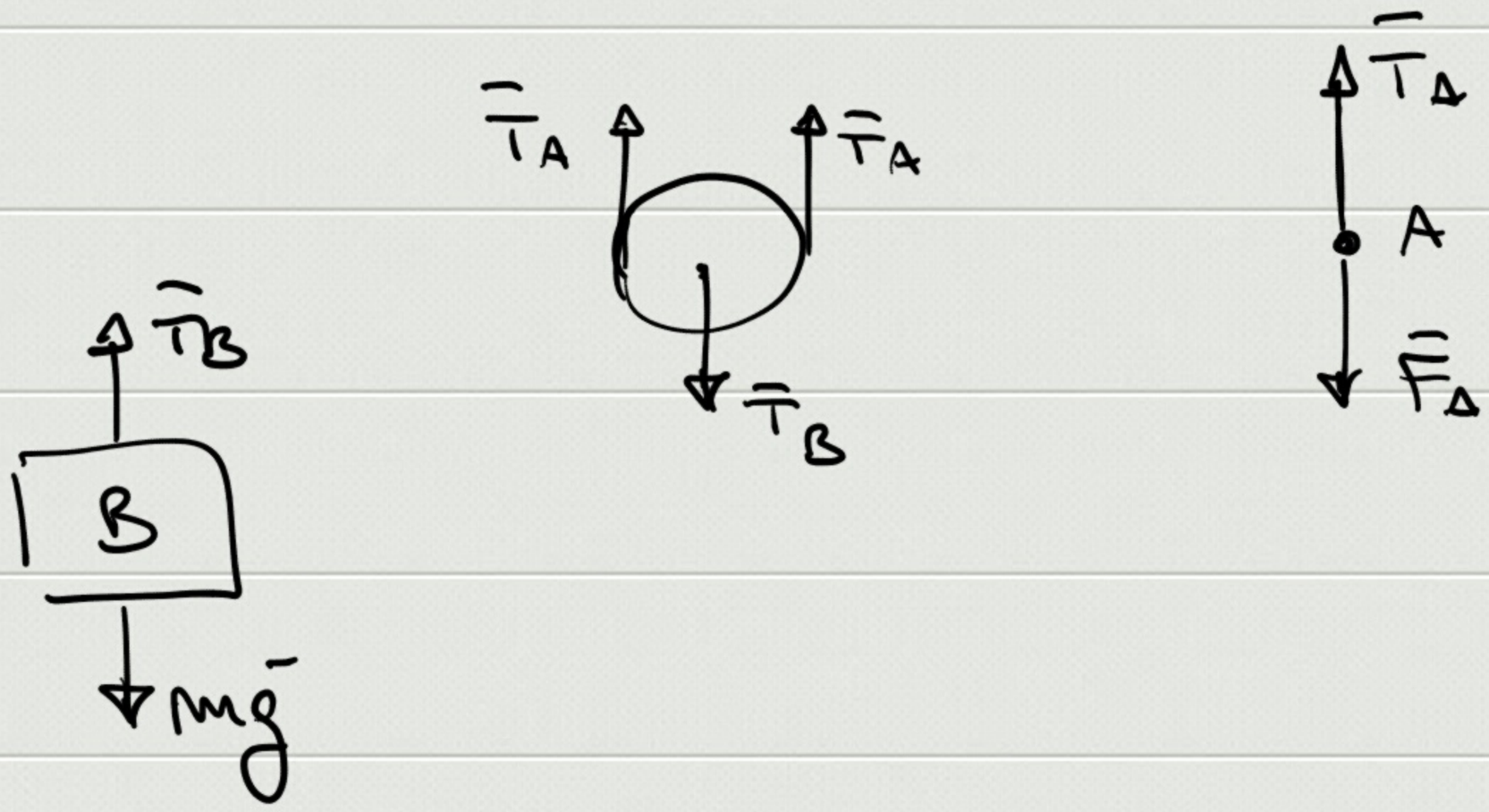


$m_B = m = 10\,kg$

$F_A = ?$ statico



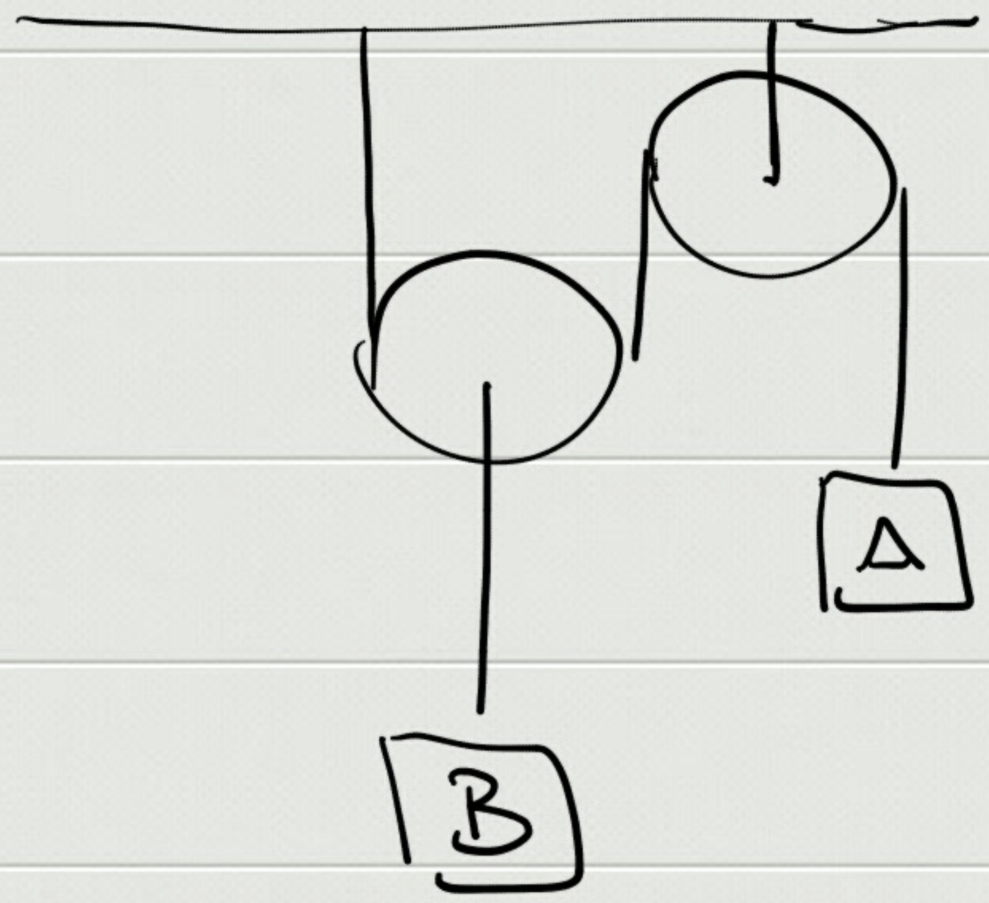
$T_B = mg$

$T_B = 2T_A$

$\Rightarrow T_A = \frac{T_B}{2} = \frac{1}{2}mg$

$F_A = T_A = \frac{1}{2}mg =$

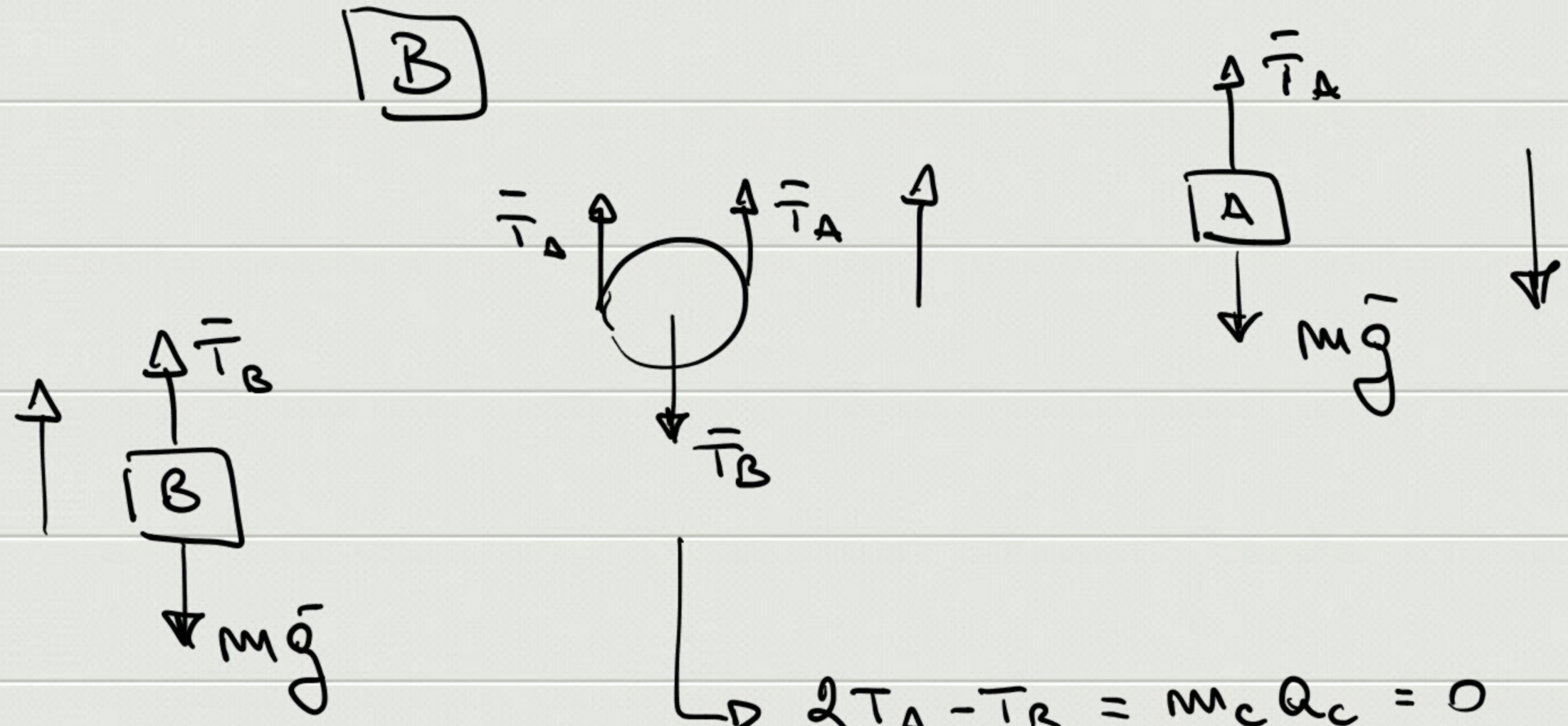
$= 49\ N$



$m_A = m_B = m = 10\ kg$

$a_A = ?$

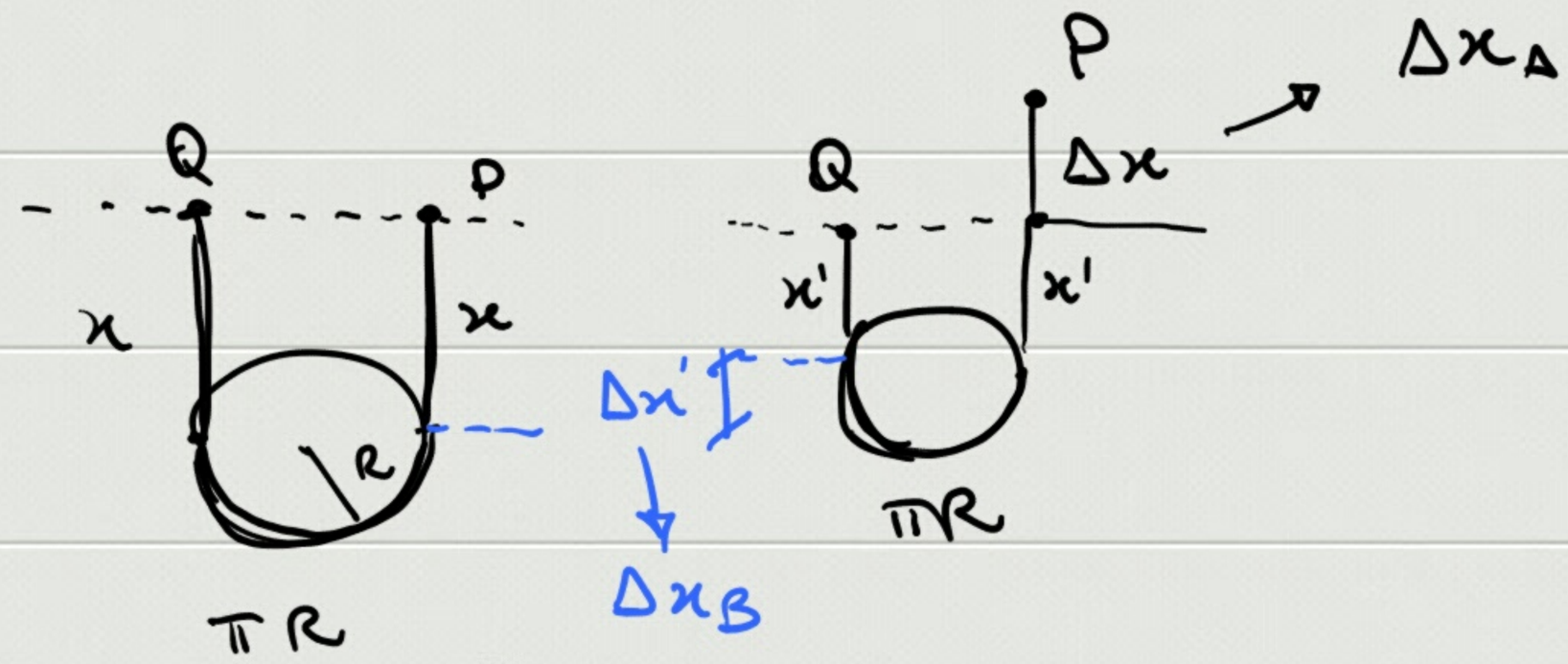
$a_B = ?$



$$2T_A - T_B = \underset{\sim 0}{\underset{\shortparallel}{m_c a_c}} = 0$$

$$\boxed{T_A = \frac{T_B}{2}}$$

$$\begin{cases} T_B - mg = m a_B \\ mg - T_A = m a_A \end{cases}$$



$$QP = 2x + \cancel{\pi R} = 2x' + \cancel{\pi R} + \Delta x$$

$$\Rightarrow x - x' = \frac{1}{2}\Delta x \Rightarrow \Delta x_B = \frac{1}{2}\Delta x_A$$

$$\Rightarrow v_B = \frac{1}{2} v_A \Rightarrow \boxed{a_B = \frac{1}{2} a_A}$$

$$\begin{cases} T_B - mg = m a_B \\ mg - T_A = m a_A \end{cases} \Rightarrow \begin{cases} 2T_A - mg = m\dfrac{a_A}{2} \\ \left[mg - T_A = m a_A\right] \times 2 \end{cases}$$

$$\cancel{m}g = \frac{5}{2}\cancel{m} a_A$$

$$\Rightarrow \boxed{a_A = \frac{2}{5} g} \qquad \boxed{a_B = \frac{1}{5} g}$$